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Ionic Liquid Lubrication of Stainless Steel: Friction is Inversely Correlated with Interfacial Liquid Nanostructure

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ABSTRACT

Atomic force microscopy (AFM) is used to measure friction of a stainless steel surface lubricated with six different ionic liquids (ILs). The measurements reveal that, contrary to expectations, friction and the degree of interfacial ionic liquid nanostructure are inversely correlated. Six ILs were confined between a sharp Si tip and a stainless steel surface, and the friction force was measured as a function of normal load. Trihexyl(tetradecyl)phosphonium bis(trifluoromethane)sulfonamide ($P_{6,6,6,14}$ TFSI) and trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate ($P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$) are the most effective lubricants, reducing lateral forces and friction coefficients by up to 3 times compared with the imidazolium-based ILs and the shorter-chained $P_{4,4,4,1}$ TFSI. Force – separation profiles reveal no obvious interfacial layers for $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$ and $P_{6,6,6,14}$ TFSI, in contrast to extensive layering for the other ILs. The bulkiness of the $[P_{6,6,6,14}]^+$ cation weakens the electrostatic interaction with the anion, resulting in weaker structure near the surface. The sliding AFM tip expels the weak near surface structure with minimal energy loss (low friction), whereas for the strongly structured ILs, more energy is dissipated by breaking the stronger interlayer bonds. The inverse correlation between friction and IL structure can be used to design weakly structured ILs which are excellent lubricants.

KEYWORDS: lubricants, friction, interfacial structure, atomic force microscopy, friction force microscopy, friction mechanism

INTRODUCTION

Lubricants reduce friction and wear between sliding surfaces. Under low pressures, viscous fluid dynamic forces separate the surfaces and prevent physical contact.[1] At high pressures, the bulk of the lubricant is squeezed out, leaving only a thin layer of lubricant (~ one molecule thick) between the sliding surfaces. This is known as boundary lubrication, and the layer of lubricant molecules the boundary layer. In conventional lubricants, the boundary layer is often formed by surfactant or polymer base oil additives, which physically adsorb to the surfaces.[1-2]

Ionic liquids (ILs) are pure salts that are liquid at low temperatures. ILs have attracted considerable interest[3-7] as potential next-generation lubricants due to their low vapor pressure,[8] high thermal stability,[9-10] good thermal conductivity,[11] and their charged nature means they generally bind strongly to solid surfaces and therefore resist squeeze out at high force. Regardless of the charge of the substrate, IL lubricity depends on both the cation and anion species.[4, 12-14] This affords the opportunity to tailor the ion structures to best lubricate a surface of interest, but as there are an estimated 10^{18} possible ILs,[15] it is not feasible to test each and every possible IL, or even IL subsets, on a surface of interest. However, understanding the IL lubrication mechanisms will enable ion structural motifs to be targeted, and fine-tuned at the molecular scale to maximize performance.

All liquids form interfacial layers at solid surfaces, known as solvation layers, with order that decreases with distance from the surface.[16-18] The high cohesive forces inherent to ILs mean they often form well defined interfacial layers.[19-22] In the IL bulk, cation and anion charge groups associate together and solvophobically exclude cation alkyl chains once a minimum threshold is reached (C_2 for protic ILs and C_4 for aprotic ILs).[23] This causes alkyl chains to cluster together leading to polar and apolar domains that percolate through the IL bulk in a sponge-like structure.[24] The presence of a solid surface aligns these domains into interfacial layers, which decay to the bulk sponge over a few nanometers, but the preferential orientation of ions is retained in the interfacial structure.[25] This distinguishes IL interfacial structure from molecular liquid solvation layers, where molecules are not significantly orientated. The number of layers that exist before the interfacial nanostructure to decays to the bulk morphology varies with the IL structure,[26-27] surface potential,[26, 28] temperature,[29-30] surface roughness,[31] and added electrolyte concentration[32-33].

We have previously investigated the effect of varying ion structures on lubricity for protic ILs on mica. Protic ILs are not practical for real world lubrication for a variety of reasons, including their water sensitivity and reactivity. These systems were explored because their bulk[24] and interfacial[19] structures were well understood, which enabled elucidation of nanostructure / lubricity relationships. For example, propylammonium nitrate (PAN), which forms several interfacial layers, was a less effective lubricant than dimethylethylammonium formate (DMEAF), which forms only a single interfacial layer.[34] This was attributed to reduced cohesive interactions at the DMEAF interface meaning that near surface ions could be more easily displaced from the interstitial space between the sliding surfaces. This result is in stark contrast to those for alkanes, alcohols and self-assembled monolayers (SAMs), where longer alkyl chains increase the strength of lateral interactions between boundary layer molecules, reducing energy dissipation and lowering friction.[35-37]

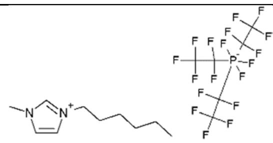
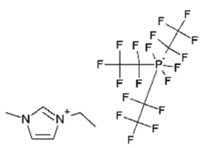
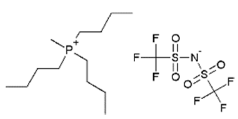
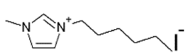
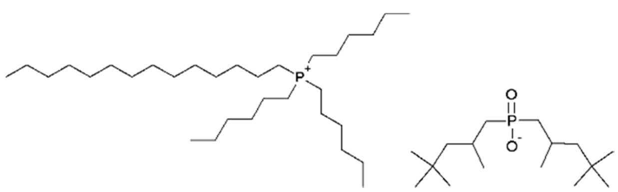
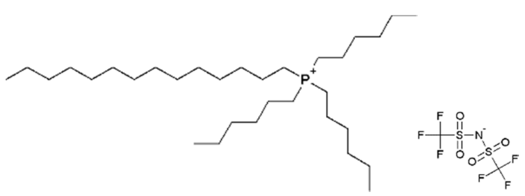
It is unclear whether these results for protic ILs on mica translate to aprotic IL lubrication of commercially relevant surfaces. In this paper, we address this knowledge gap by studying the lubrication of stainless steel using six aprotic ILs. Notably, aprotic ILs can be hydrophobic (moisture resistant) and unreactive,[38] and therefore potentially applied in real world settings. To our knowledge, this is the first nanotribology study of stainless steel with ILs. Atomic force microscopy (AFM) is used to measure both normal and lateral forces, to establish relationships between IL interfacial structure and lubricity, and strategies for minimizing energy dissipation.

EXPERIMENTAL

The ILs 1-ethyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate (EMIM FAP) (purity ≥ 98%) and 1-hexyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate (HMIM FAP) were purchased from Merck (purity ≥ 99%). All other ILs, 1-hexyl-3-methylimidazolium iodide (HMIM I) (purity > 98%), tributyl(methyl)phosphonium bis(trifluoromethane)sulfonamide ($P_{4,4,4,1}$ TFSI) (purity > 98%), trihexyl(tetradecyl)phosphonium bis(trifluoromethane)sulfonamide ($P_{6,6,6,14}$ TFSI) (purity > 98%), and trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate ($P_{6,6,6,14}$ $(^iC_8)_2PO_2$) (purity > 95%), were supplied by Iolitec and used as received. The IL structures and key physical parameters are presented in Table 1.

An AISI 316 stainless steel-coated quartz crystal microbalance (QCM) sensor (KSV Instruments, Helsinki) was used as the substrate in the AFM experiments. The roughness of the stainless steel was measured by AFM. For a 500 nm × 500 nm scan (Figure S1), the RMS roughness was 0.89 nm.

Table 1. The molecular structures, ion pair diameter and viscosity for the ILs used in this study.

Ionic Liquid	Molecular Structure	*d* (nm) [a, b]	η (mPa · s) [b]
HMIM FAP		0.87[39]	88[39]
EMIM FAP		0.81[40]	59[41]
$P_{4,4,4,1}$ TFSI		0.87[42]	207[42]
HMIM I		0.71[43]	1730[c]
$P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$		1.13[44]	1402[44]
$P_{6,6,6,14}$ TFSI		1.05[44]	337[44]

[a] The ion pair diameter was calculated from the density assuming cubic packing.

[b] Data is for 25 °C.

[c] From the manufacturer.

Normal and friction force measurements were performed using a Bruker Multimode 8 AFM with an EV scanner in contact mode. Sharp Si tips with a spring constant of 1.0 ± 0.2 N/m as measured by the thermal tune method and a nominal tip radius of 8 nm (HQ:CSC37/AL BS, Mikromasch) were used for this study. An AFM fluid cell (Bruker) was used to perform the experiments. Before each experiment the tip and the cell were carefully rinsed with ethanol and Milli-Q water and then dried with nitrogen. The tip was subsequently irradiated with ultraviolet light for at least 30 min to remove any remaining organic matter. Friction measurements were performed using a scan size of 100 nm at a scan speed of 6.5 μm s^{-1} while the normal load was increased from 0 to 100 nN. The lateral deflection signal (i.e. cantilever twist) was converted to lateral force using a customised function produced in MATLAB R2014b. The friction coefficient (μ) of each data set was obtained using Amontons' Law, $F_L = \mu F_N + F_L(0)$, where μ is extracted from the gradient of the lateral force vs. normal load in the linear region and $F_L(0)$ is the lateral force at zero normal load. At least five runs of both increasing and decreasing load were recorded in each experiment, and at least three experiments were performed for each IL. As the trends for decreasing load were the same, only increasing load for a single run is reported here for clarity.

RESULTS AND DISCUSSION

Most previous studies of IL nanotribology have used very (or atomically) smooth substrates such mica,[45-50] graphite,[51] gold,[52-53] and silica.[54-55] While these smooth, well defined, surfaces are critical for teasing out fundamental understanding of IL lubrication mechanisms, they are not as industrially relevant as stainless steel.[56-57] Therefore, this study uses a stainless steel coated QCM sensor which is smooth enough (RMS roughness = 0.89 nm) as the substrate for force measurements. Stainless steel is an iron alloy containing ~ 10 – 20% chromium. Surface chromium readily reacts with oxygen in air to form a passivating layer of chromium oxide.[58] It is this surface that is lubricated.

AFM lateral (friction) force data for six different ILs lubricating a sharp Si tip sliding on stainless steel as a function of load are presented in Fig 1. When the tip and surface are well-separated, the lateral forces measured are negligible. At approximately zero normal load, attractive van der Waals forces cause the tip and the surface to 'jump' into an adhesive contact,[52, 54] and the lateral force increases sharply to between 8 nN and 30 nN depending on the IL. In the absence liquid (tip on stainless steel in air) the initial jump in lateral force is much higher, at ~150 nN (Supporting Information Figure S2). This shows these ILs effectively screen long range van de Waals interactions, as has been noted previously in other systems.[52, 54]

After the initial jump the lateral forces increase linearly with normal load up to 100 nN, which was the highest load measured. The friction coefficient (μ) is gradient of this linear increase, and therefore is independent of the adhesion, as described by Amontons' law. The friction coefficients for each IL are presented in Table 2. The force versus distance curves presented in Figure 2 show for forces greater than 2 nN it is likely the tip and stainless steel are separated only by the layer of ions bound to the stainless steel. Ions will also adsorb to the silica tip, but as the tip is rougher (in the vicinity of the measurement asperity) than the stainless steel, which weakens lateral interactions between ions, meaning they are displaced at low force. Non-linear or 'quantized' friction regimes[45-46] could operate at low loads (0 and 5 nN), but the focus of this paper is on high loads (i.e. Hertzian contact pressures ranging from 0 to 9.8 GPa) which are more relevant to real world systems.

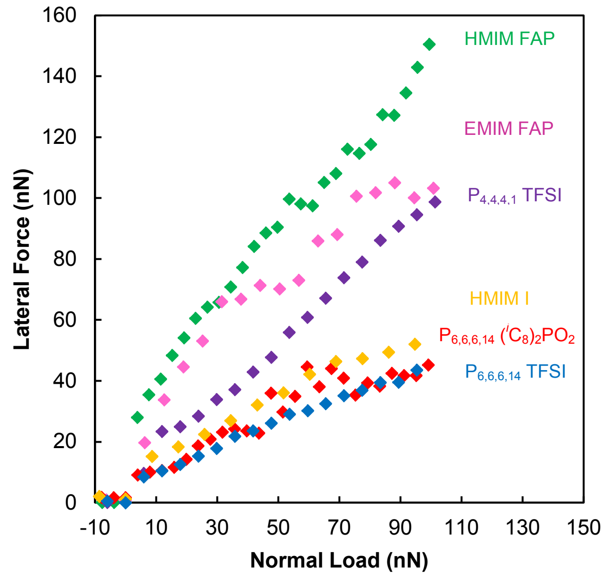


Figure 1. Lateral force versus normal load in ionic liquid for a sharp silicon AFM tip sliding on a stainless steel surface over 100 nm at 6 μm s^{-1}.

Table 2. Friction coefficient μ. Errors are ±0.10.

	μ
Air	0.88
HMIM FAP	1.16
EMIM FAP	0.84
$P_{4,4,4,1}$ TFSI	0.93
HMIM I	0.46
$P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$	0.40
$P_{6,6,6,14}$ TFSI	0.39

Lateral forces in Figure 1 increase in the order $P_{6,6,6,14}$ TFSI ≈ $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$ < HMIM I < $P_{4,4,4,1}$ TFSI < EMIM FAP < HMIM FAP. The order of the friction coefficients (Table 2) in the ILs is the same, with the exception that the friction coefficient for EMIM FAP is slightly lower than for $P_{4,4,4,1}$ TFSI. The two $P_{6,6,6,14}$ ILs are the most effective lubricants for stainless steel. The chromium oxide layer on the surface of the stainless steel is slightly negatively charged,[59] which means that the IL cation charge groups will preferentially adsorb to the surface. The data sets for $P_{6,6,6,14}$ TFSI and $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$ almost overlap, which is consistent with a cation-rich boundary layer. $[P_{6,6,6,14}]^+$ has a surfactant-like structure, which could lead to high surface activity and strong adsorption. HMIM I is nearly as lubricating as $P_{6,6,6,14}$ TFSI and $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$, and much more lubricating than HMIM FAP, revealing a clear role for the anion for $[HMIM]^+$ ILs, which is not obvious for $[P_{6,6,6,14}]^+$ ILs. This is likely a consequence of the $[P_{6,6,6,14}]^+$ charge group being much bulkier than that of $[HMIM]^+$. The bulkiness of the $[P_{6,6,6,14}]^+$ charge group prevents anions from being incorporated into the boundary layer, but the unhindered $[HMIM]^+$ leaves space available for anion inclusion. The higher lubricity of HMIM I compared with HMIM FAP is attributed to the smaller size iodide compared with FAP leading to neat packing in the boundary layer. A well-formed boundary layer reduces energy dissipation pathways and lowers friction. Iodide based lubricants are probably not practical for real world systems due to reactivity issues, but these results indicate that small anions may be useful for stainless steel.

Several previous articles – including our own earlier work – have suggested that well formed interfacial IL nanostructure is desirable for lubrication. The assumptions are that at low loads interfacial layers will slide smoothly over one another (akin to lubrication by graphite) while in the high load regime the same forces that lead to well-formed nanostructure result in strong interactions between adsorbed ions, resulting in a robust boundary layer.[12, 45, 54] To test these assumptions, knowledge of the strength of interfacial structure is required. AFM force – separation profiles measured for EMIM FAP, HMIM FAP, HMIM I, $P_{4,4,4,1}$ TFSI, $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$, and $P_{6,6,6,14}$ TFSI confined between a sharp silicon AFM tip (r = 8 nm) and a stainless steel surface are presented in Figure 2. At least 30 normal force curves were measured for each IL during each experiment, and these experiments were repeated at least three times. Representative force curves from all experiments are presented in Figure 2. Normal forces up to 80 nN were measured, but for all the ILs tested no further layers were detected at normal forces greater than 2 nN.

For IL force – distance curves, the number of steps, and the push through forces, indicate the number of layers, and the strength of cohesive interactions between ions in layers, respectively. ILs with strongly structured interfaces have many steps with high push-through forces, whereas ILs with weak interfacial structure have few or no steps.[19] In Figure 2, at large separations between the tip and the stainless steel (> 5 nm) no force is measured; any structure in the bulk of the ILs is too weak to deflect the tip, and is readily displaced as the tip approached. At smaller separations (< 5 nm) the steps in the data correspond to layers of IL.

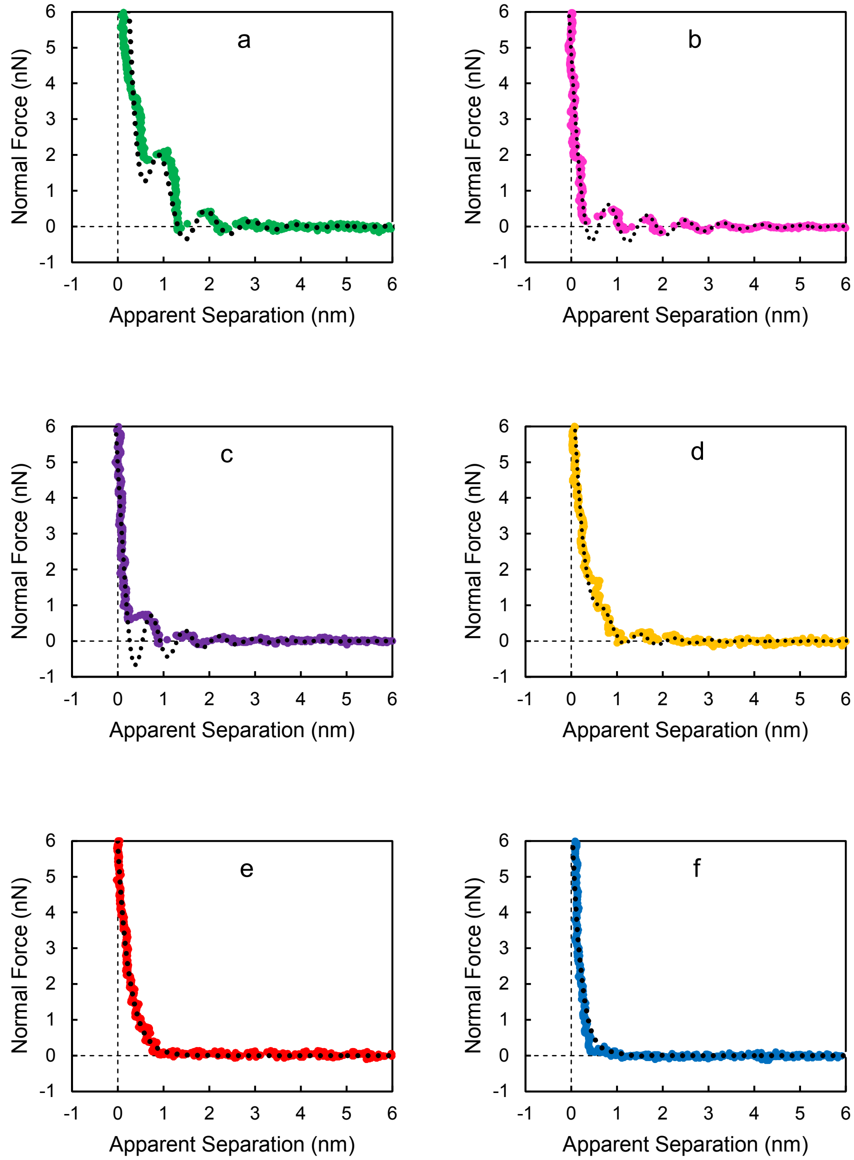


Figure 2. Typical force versus apparent separation profiles in different ionic liquids confined between a sharp silicon AFM tip and a stainless steel surface. (a) HMIM FAP, (b) EMIM FAP, (c) $P_{4,4,4,1}$ TFSI, (d) HMIM I, (e) $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$, (f) $P_{6,6,6,14}$ TFSI. The scan size was 30 nm, and the scan rate was 0.2 Hz. The dotted line is the fit using Equation 1.

The number of steps in the force curve and rupture forces varies considerably with IL. While there are clear steps for HMIM FAP (Figure 2a), EMIM FAP (Figure 2b), P_{4441} TFSI (Figure 2c), and HMIM I (Figure 2d), no steps were measured for $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$ (Figure 2e) and $P_{6,6,6,14}$ TFSI (Figure 2f). For $P_{6,6,6,14}$ $(^{i}C_8)_2PO_2$ (Figure 2e) and $P_{6,6,6,14}$ TFSI, only a slight compression over a distance of less than 1 nm is measured, which is consistent with the compression of solution facing C_{14} alkyl chains, but no other interfacial structure (within the resolution of the measurement). That is, contrary to our previous expectations, the ILs with the weakest interfacial structure are the best performing lubricants.

The force required to rupture a layer must match the sum of the cohesive forces of the ions that are displaced. To better quantify the strength of this interaction, the force curve data was fit using a model developed by Hoth et al.,[60] which describes the interaction potential $v_{int}(g)$ approaching the surface as

$$v_{int}(\mathrm{g})=v_0 \cos\left(\frac{2\pi g}{d}\right)\exp\left(-\frac{g}{\lambda_0}\right)+\mathrm{v}_1\exp\left(-\frac{g}{\lambda_1}\right) \qquad (1)$$

where v_i and λ_i (with i = 0, 1) are surface energies and correlation lengths respectively, and d represents the width of the IL ion pair. A full description of the equation can be found in Hoth et al.,[60] but briefly the model interprets the IL force – distance data as a Hertzian contact with hard wall repulsion and finite range potential. The first term of the equation models the sinusoidal decay of the steps in the force curves with distance from the surface, while the second term accounts for the exponential repulsion immediately adjacent to surface. The fitted parameters are shown in Table 2. The fit assumes that at zero apparent separation there is still an ion pair separating the tip and the surface. The AFM force – distance profile does not capture all the minima in the potential because the cantilever has a finite spring constant, and instead jumps to the next oscillation (layer).

The fits to the force data using Equation 1 and the parameters in Table 3 are indicated by the dashed lines in Figure 2. The trends in surface energy in Table 2 are consistent with our interpretations of step number/push through force. The near surface energies v_0 increase in the order $P_{6,6,6,14}$ TFSI ≈ $P_{6,6,6,14}$ $(^iC_8)_2PO_2$ < HMIM I < EMIM FAP < $P_{4,4,4,1}$ TFSI < HMIM FAP. This is the same order as the increase in lateral forces and friction coefficients from the friction experiments (cf. Table 2), aside with the position of EMIM FAP. However, inspection of the force curve for EMIM FAP in Figure 2 reveals a very small step at close separations that the model does not capture, which could account for this difference; this small step was consistent between repeat experiments. In any case, the key result is that $P_{6,6,6,14}$ $(^iC_8)_2PO_2$ and $P_{6,6,6,14}$ TFSI, with little or no forces between layers ($v_0 \leq 0.01$), reduce friction most effectively ($\mu \leq 0.4$). In contrast, ILs with stronger attractive forces between layers, such as HMIM FAP and $P_{4,4,4,1}$ TFSI (v_0 = 0.14 and 0.07 J m^{-2} respectively), are worse at reducing friction (μ = 1.16 and 0.93). This is consistent with the earlier observation that ILs with the weakest interfacial structure are the best performing lubricants.

Table 3. Force curve fitting parameters for Equation 1.

IL	d (nm)	v_0 (J m^{-2})	λ_0 (nm)	v_1 (J m^{-2})	λ_1 (nm)
HMIM FAP	0.98	0.14	1	0.33	0.38
EMIM FAP	0.83	0.03	1.4	0.6	0.18
$P_{4,4,4,1}$ TFSI	0.74	0.07	0.9	0.7	0.16
HMIM I	0.76	0.02	1.2	0.2	0.29
$P_{6,6,6,14}$ $(^iC_8)_2PO_2$	1.13	0	0.1	0.9	0.26
$P_{6,6,6,14}$ TFSI	1.00	0.01	0.7	2.0	0.2

In the boundary regime energy is dissipated mainly via: (i) expulsion of the near surface IL multilayers from the space between the tip and the surface and (ii) deformations and rotations of ions in the boundary layer.[34, 61] When the friction data (Figure 1) and force curve data and fits (Figure 2) are considered, together a clearer picture of how *(i) expulsion of the near surface IL multilayers* leads to low friction for $P_{6,6,6,14}$ ILs emerges. In the boundary layer, the $P_{6,6,6,14}$ cation is adsorbed with its charge preferentially orientated towards the chromium oxide layer. This presents cation alkyl chains to solution, which may be expected to solvophobically associate with solution cations. However, the strong steric hindrance of the cation charge group means electrostatic interactions between this cation and anions are weak compared with the other ILs. Weakening of electrostatic interactions reduces the solvophobic tendency[23] for cation alkyl chains to associate, resulting in weak interfacial nanostructure. This means the surface bound cations interact weakly with near surface anions (electrostatically) and cations (solvophobically), and the *near* surface layers are readily displaced. The movement of near surface ions is the same, parallel to the plane of the surface, under compression of the AFM tip in a normal force experiment or by the lateral movement of the AFM tip in a friction experiment. This relationship accounts for the observation that lateral force is inversely correlated with interfacial structure. The bulk structure of these ILs has previously been studied by the Castner group and the Margulis group.[62-63] These studies show a disordered, more homogenous nanostructure compared with other ILs, consistent with the results in this work.

By way of contrast, the short alkyl chains on the $P_{4,4,4,1}$ cation means electrostatic interactions with the anion are stronger. The asymmetry of the cation prevents solvophobic association of alkyl chains, but cation – anion electrostatic interactions result in at least four steps in the $P_{4,4,4,1}$ TFSI force – distance profile, compared with one weak step observed for $P_{6,6,6,14}$ TFSI. Stronger electrostatic interactions, and solvophobic association for cations with alkyl chains C_4 or longer, mean the other ILs have stronger bulk and interfacial structure, leading to stepped force curves, cf. HMIM FAP (Figure 2a), EMIM FAP (Figure 2b), and HMIM I (Figure 2d). While energy must also be dissipated via *(ii) deformations and rotations of ions in the boundary layer* the inverse correlation between normal and lateral force over the 6 ILs implies a key role for interactions between the boundary layer and near surface ions, as discussed previously: For protic ILs with stronger interfacial structure (PAN, PAF) showed higher friction than weakly structured ILs (DMEAF, EtAN).[34]

It is noteworthy that trends in the lateral force data in Figure 1 do not match the trends in IL bulk viscosity (Table 1). For example, the viscosity of HMIM FAP is higher than EMIM FAP (88 vs 59 mPa · s), yet lateral forces in HMIM FAP are significantly higher than EMIM FAP. This means the mechanism for energy dissipation in the boundary regime is independent of bulk viscosity, which is also consistent with a previous AFM nanotribology study of protic ionic liquids on mica.[34]

CONCLUSION

The inverse correlation between friction and IL interfacial structure has been revealed using AFM. All IL cations in this study ($[HMIM]^+$, $[EMIM]^+$, $[P_{4,4,4,1}]^+$, and $[P_{6,6,6,14}]^+$) adsorb strongly to the chromium oxide layer of the stainless steel and thus friction is largely determined by the cation. The bulkiness of the $[P_{6,6,6,14}]^+$ charge group weakens the interaction between it and the anion. This means that the surface bound cations in $P_{6,6,6,14}$ ILs interact weakly with near surface anions (electrostatically), and cations (solvophobically), and the near surface layers are readily displaced. Lateral forces are up to 3 times lower in $P_{6,6,6,14}$ ILs compared with the HMIM, EMIM, and $P_{4,4,4,1}$. In these ILs, the charge group density is much higher, which means electrostatic interactions with the anion are stronger. Stronger interactions between the adsorbed boundary layer and the near surface layers require more energy to rupture them during boundary layer lubrication. This results in higher lateral forces for these ILs with more interfacial structure.

Previous work has given us a good understanding of IL molecular structure affects bulk and interfacial nanostructure. This work suggests that we can use this knowledge to design ILs with weak near surface structure which are excellent lubricants.

ASSOCIATED CONTENT

SUPPORTING INFORMATION

The Supporting Information is available free of charge on the ACS Publications website. This includes the contact mode AFM image of the stainless steel surface in air, and lateral force vs normal load measured on stainless steel in air.

AUTHOR INFORMATION

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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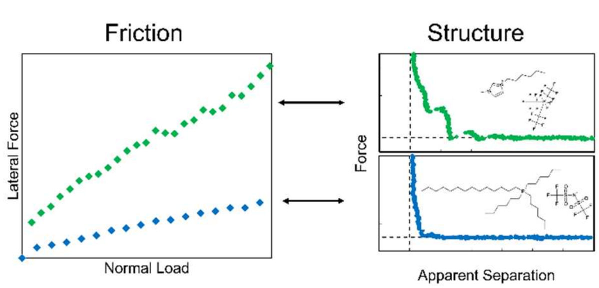

Friction in ionic liquids inversely correlates with the interfacial nanostructure. This paves the way for the design of high performance ionic liquid lubricants.